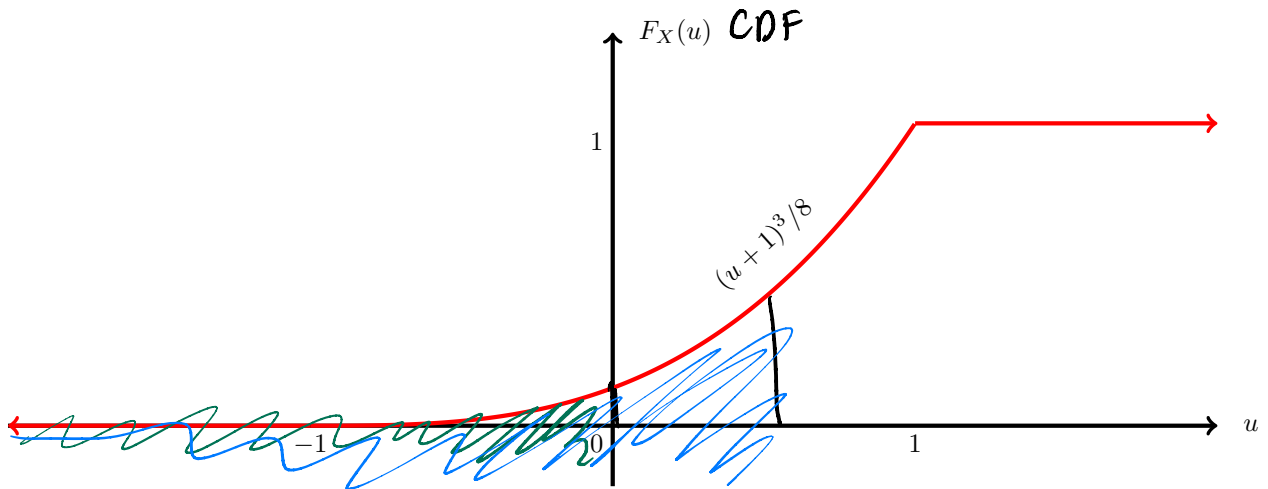


Let X be a random variable whose cumulative distribution function is $(u+1)^3/8$ on $[-1, 1]$, as shown below. What is the probability that $0 < X < 3$, given that $-2 < X < 1/2$?



(a) 19/27

(b) 8/27

(c) 1/8

(d) 4/9

(e) 5/9

(f) 1/4

(g) 3/4

(h) 1

(i) 1/9

(j) 1/3

(k) 1/2

(l) None of these

$$P\left(0 < X < 3 \mid -2 < X < \frac{1}{2}\right)$$

$$= \frac{P\left((0 < X < 3) \cap (-2 < X < \frac{1}{2})\right)}{P(-2 < X < \frac{1}{2})}$$

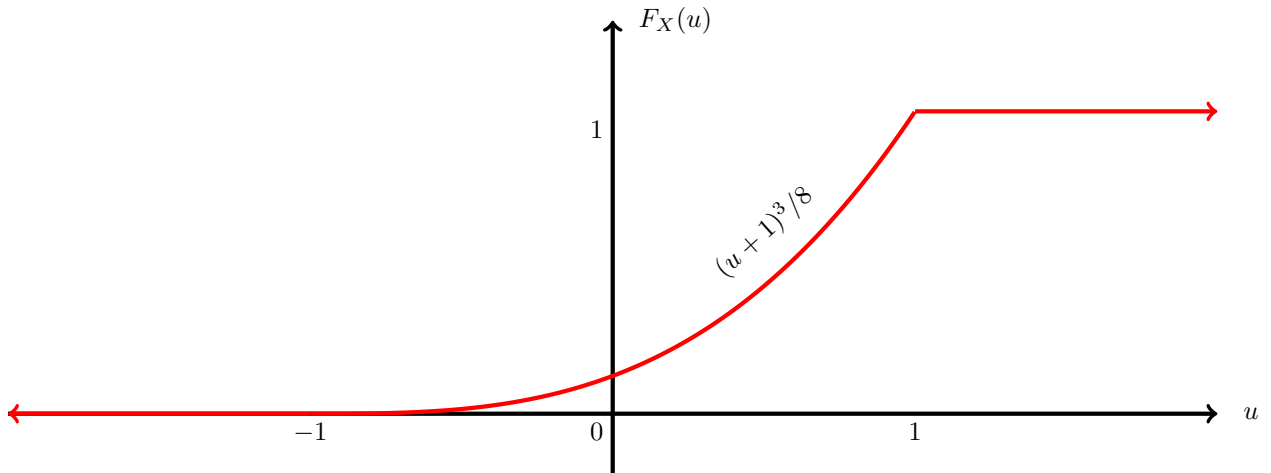
$$= \frac{P\left(0 < X < \frac{1}{2}\right)}{P(-2 < X < \frac{1}{2})}$$

$$= \frac{F_X\left(\frac{1}{2}\right) - F_X(0)}{F_X\left(\frac{1}{2}\right) - F_X(-2)}$$

$$= \frac{\left(\frac{1}{2} + 1\right)^3/8 - (0+1)^3/8}{\left(\frac{1}{2} + 1\right)^3/8 - 0}$$

$$\rightarrow \text{calculator} \rightarrow = \frac{19}{27}$$

Let X be a random variable whose cumulative distribution function is $(u+1)^3/8$ on $[-1, 1]$, as shown below. What is the expected value of X ?



- (a) $1/2$
- (b) $-1/2$
- (c) $2/3$
- (d) $-2/3$
- (e) 1
- (f) -1
- (g) $1/8$
- (h) $-1/8$
- (i) $1/4$
- (j) $3/4$
- (k) $3/8$
- (l) None of these

$$E[X] = \int_{-\infty}^{\infty} u f_X(u) du$$

$$\begin{aligned} f_X(u) &= \frac{d}{du} F_X(u) \\ &= \frac{d}{du} \left(\frac{(u+1)^3}{8} \right) \\ &= \frac{3(u+1)^2}{8} \end{aligned}$$

$$E[X] = \int_{-1}^1 u \frac{3(u+1)^2}{8} du$$

$$= \frac{3}{8} \int_{-1}^1 u(u+1)^2 du$$

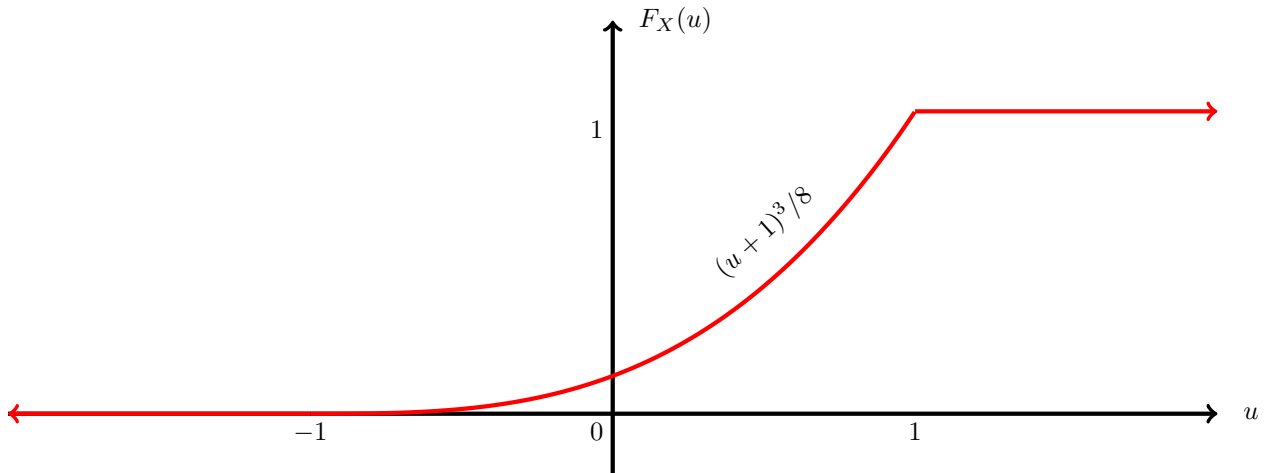
$v = u+1$
 $dv = du$

$@ u=1, v=2$
 $@ u=-1, v=0$

$$= \frac{3}{8} \int_0^2 (v-1)v^2 dv$$

$$= \frac{3}{8} \int_0^2 (v^3 - v^2) dv$$

Let X be a random variable whose cumulative distribution function is $(u + 1)^3/8$ on $[-1, 1]$, as shown below. What is the expected value of X ?



- (a) $1/2$
- (b) $-1/2$
- (c) $2/3$
- (d) $-2/3$
- (e) 1
- (f) -1
- (g) $1/8$
- (h) $-1/8$
- (i) $1/4$
- (j) $3/4$
- (k) $3/8$
- (l) None of these

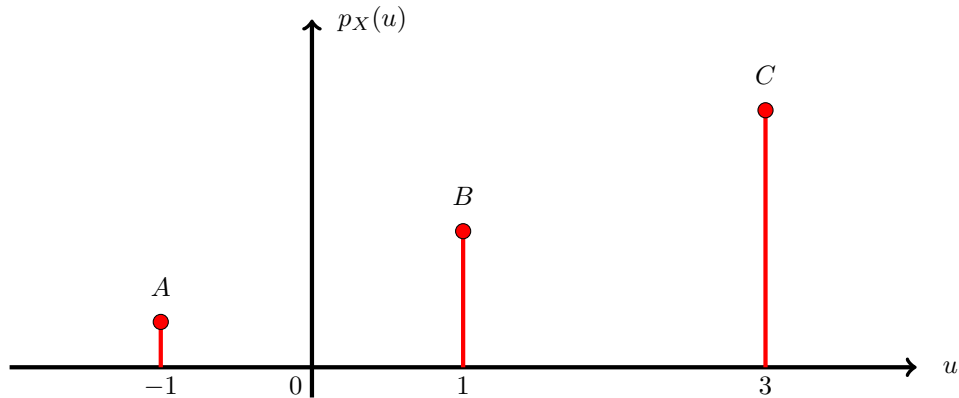
$$= \frac{3}{8} \left[\frac{1}{4} v^4 - \frac{1}{3} v^3 \right]_0^2$$

$$= \frac{3}{8} \left(4 - \frac{8}{3} \right)$$

$$= \frac{3}{8} \left(\frac{12}{3} - \frac{8}{3} \right)$$

$$= \frac{12}{24} = \frac{1}{2}$$

Let X be a discrete random variable whose probability mass function is shown below. If the expected value of X is $5/2$, and the probability that X^2 is larger than $2(3X - 4)$ is $1/5$, then what is B ?



$$E[X] = \frac{5}{2}$$

(a) $3/20$

(b) $3/10$

(c) $1/10$

(d) $1/4$

(e) $1/8$

(f) $7/10$

(g) $5/20$

(h) $3/4$

(i) $3/8$

(j) $7/20$

(k) $1/2$

(l) None of these

$$\frac{5}{2} = -A + B + 3C$$

$$P(X^2 > 2(3X - 4)) = \frac{1}{5}$$

$$P(X^2 > 6X - 8) = \frac{1}{5}$$

$$P(X^2 - 6X + 8 > 0) = \frac{1}{5}$$

$$P((X-4)(X-2) > 0) = \frac{1}{5}$$

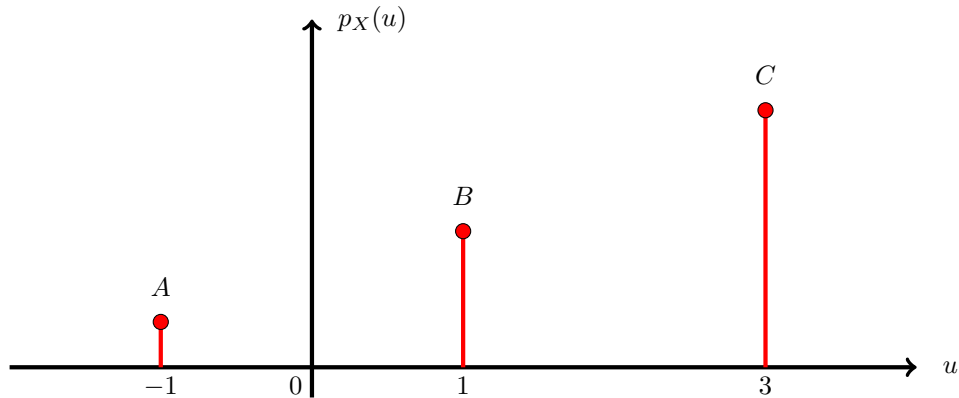
$$\hookrightarrow \text{valid @ } X \in \{-1, 1\}$$

$$P(X = -1) + P(X = 1) = \frac{1}{5}$$

$$A + B = \frac{1}{5}$$

$$A + B + C = 1$$

Let X be a discrete random variable whose probability mass function is shown below. If the expected value of X is $5/2$, and the probability that X^2 is larger than $2(3X - 4)$ is $1/5$, then what is B ?



system 3 eq

(a) $3/20$

(b) $3/10$

(c) $1/10$

(d) $1/4$

(e) $1/8$

(f) $7/10$

(g) $5/20$

(h) $3/4$

(i) $3/8$

(j) $7/20$

(k) $1/2$

(l) None of these

$$\frac{5}{2} = -A + B + 3C$$

$$A + B = \frac{1}{5}$$

$$\begin{array}{r} - \\ A + B + C = 1 \\ \hline 0 + 0 - C = -\frac{4}{5} \end{array}$$

$$\rightarrow C = \frac{4}{5}$$

$$A + B = \frac{1}{5}$$

$$+ \quad -A + B = \frac{5}{2} - 3\left(\frac{4}{5}\right)$$

$$0 + 2B = \frac{3}{10}$$

$$\rightarrow B = \frac{3}{20}$$